

Case Study

The 2018 Aadhaar

Data Leak

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Course: ... [2025-26]

Agenda



Introduction & Background of the Case



The Cyber Forensic Investigation Process



Evidence Collection and Analysis



Challenges Faced in the Investigation



Outcome and Preventive Measures



Conclusion

Introduction & Background of the Case

THE ₹500 DATA BREACH



1 BILLION CITIZENS EXPOSED



INVESTIGATION: JAN 2018



FULL DATABASE ACCESS SOLD

The Cyber Forensic Investigation Process

FORENSIC WORKFLOW

IDENTIFY



Finding the vulnerability

ANALYZE



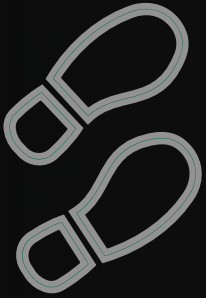
Reviewing server logs & traffic



The Cyber Forensic Investigation Process

FORENSIC WORKFLOW

TRACE



Following the digital trail

CONTAIN



Closing the security gap



Evidence Collection and Analysis

EVIDENCE COLLECTED

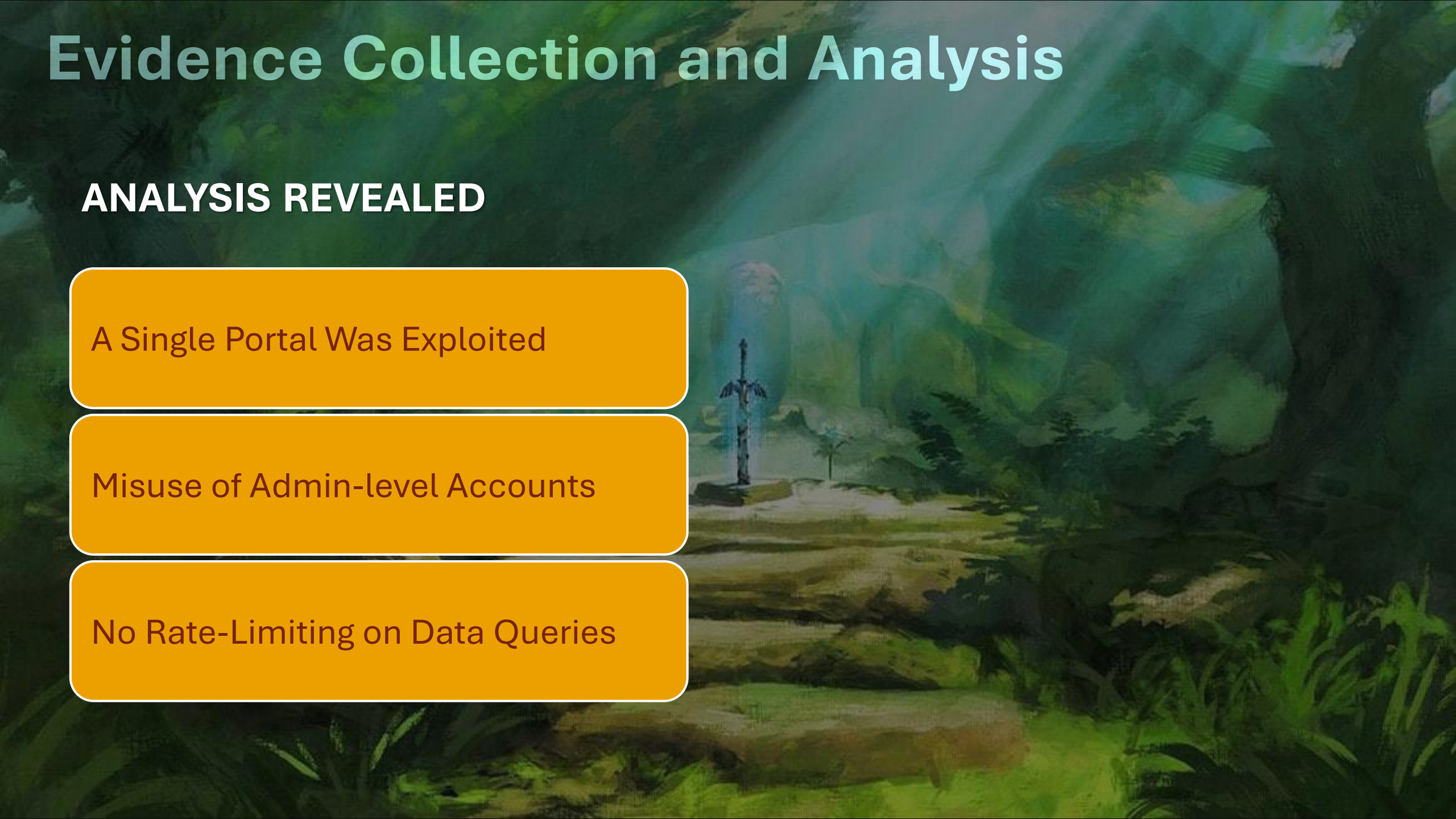
Server &
Gateway
Logs

Compromised
User
Credentials

Unauthorized IP
Addresses



Evidence Collection and Analysis

The background of the slide is a dark, atmospheric illustration of a jungle scene. A stone path leads from the foreground into the distance, where a small structure with a cross-like monument stands. The scene is dimly lit, with light filtering through the dense foliage and trees, creating a mysterious and somewhat ominous mood.

ANALYSIS REVEALED

A Single Portal Was Exploited

Misuse of Admin-level Accounts

No Rate-Limiting on Data Queries

Challenges Faced During Investigation

- **MASSIVE DATASCALE** (Over 1 billion records to protect and analyze)
- **OFFICIAL DENIAL** (Initial refusal by authorities to acknowledge a breach)
- **ANONYMOUS SELLERS** (Tracing perpetrators on encrypted platforms like WhatsApp)
- **"SHOOT THE MESSENGER"** (Legal action was taken against the reporting journalist)



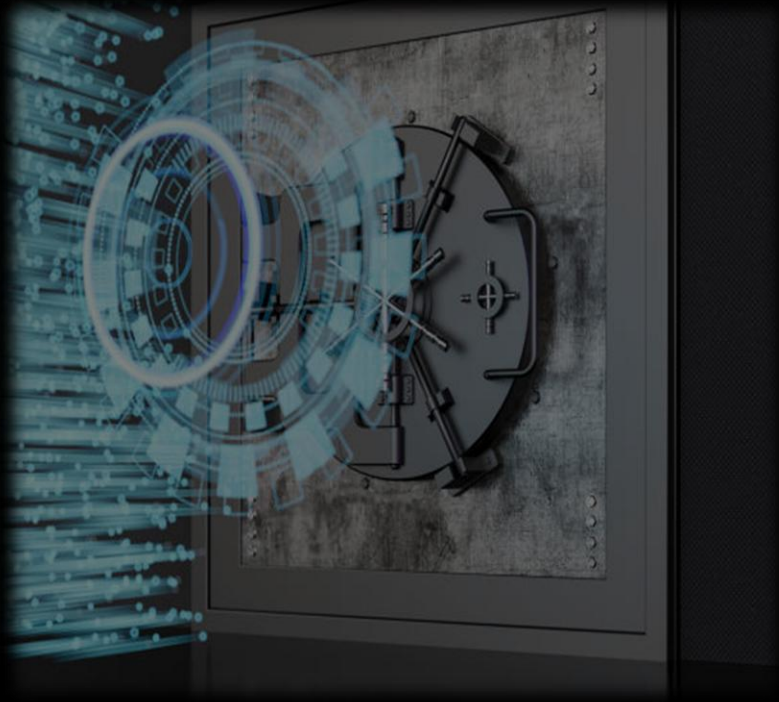
Outcome and Preventive Measures



CASE OUTCOME

- Several Arrests Made
- FIR Filed Against Journalist
- Increased Public Scrutiny

Outcome and Preventive Measures



Virtual ID (VID) Introduced
Hardened Access Protocols
Aadhaar Data Masking

Conclusion



- ❑ The 2018 Aadhaar data leak was not a sophisticated hack, but a critical failure of **access control management**, highlighting a simple but devastating vulnerability.
- ❑ The investigation demonstrated the immense challenge of digital forensics when dealing with **massive data volumes**, official denial, and anonymous perpetrators.
- ❑ This incident served as a crucial wake-up call for India, igniting a national debate on data privacy and the security of critical digital infrastructure.
- ❑ While the breach was severe, the subsequent implementation of **preventive measures** like Virtual IDs and data masking represents a significant step towards a more secure system.

Q&A

Thanking You



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